Nosocomial infection

Hospital acquired infection are also known as a nosocomial infection. It is acquired in a hospital or other health care facility. Infection is spread to the susceptible patient in the clinical setting by various means, one of them being air droplets. The infection can originate from another infected patient, staff, or in some cases, the source of the infection cannot be determined. The most common pathogens that cause nosocomial infections are *Staphylococcus aureus, Pseudomonas aeruginosa,* and *Escherichia coli.* One of the common nosocomial infections is respiratory pneumonia.

9.2.6 Enumeration of Microorganisms in Air

There are several methods adopted to enumerate microorganisms in air. The most important methods are solid impingement and liquid impingement, filtration, sedimentation, centrifugation electrostatic precipitation. Many tools have been designed for the collection of air samples. Choosing an appropriate sampling device is based on many factors, such as availability, cost, volume of air to be sampled, sampling efficiency and the environmental conditions under which sampling will be conducted. One of the methods is Settle plate technique, where the microorganism carrying particles are allowed to settle onto the medium (solid impingement) for a given period of time and incubated at the optimum temperature (Figure 9.3). It works under the principle of gravitational force.



Figure 9.3: Settle plate technique showing bacterial and fungal growth

Choice of the medium depends upon the kind of microorganisms to be enumerated. For bacterial isolation Nutrient agar and for fungal isolation Sabourauds dextrose agar (SDA) can be used.

During sampling it is better to keep the plates about one meter above the ground level. Then the plates are uncovered in the selected position for the required period of time. Immediately after exposure for the given period of time the plates are closed with the lids. Then the plates are incubated for 24 hrs at 37°C for aerobic bacteria and for 2 days at room temperature (27°C) for fungi. Enumeration results are expressed as numbers of viable organisms per unit area of air colony forming unit (CFU/ mm^3).

HOTS

Can microorganisms grow in clouds?

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